

Rahul Suryakant Sakhare, Ph.D., P.E.

Curriculum Vitae

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EDUCATION

- **Doctor of Philosophy (Ph.D.)** May 2023
Purdue University, West Lafayette, United States
Civil Engineering — Transportation & Infrastructure Systems
- **Master of Technology (M.Tech.)** May 2018
Indian Institute of Technology (IIT) Madras, India
Civil Engineering (Transportation Engineering)
- **Bachelor of Technology (B.Tech.)** May 2017
Indian Institute of Technology (IIT) Madras, India
Civil Engineering (Minor: Management Studies)

PROFESSIONAL APPOINTMENTS

- **Transportation Research Engineer** (Post-doctoral equivalent) Aug 2023 – Present
Joint Transportation Research Program (JTRP)
Lyles School of Civil & Construction Engineering
Purdue University, West Lafayette, United States
- **Graduate Research Assistant** Jan 2020 – May 2023
Advisor: Prof. Darcy M. Bullock
Lyles School of Civil Engineering
Purdue University, West Lafayette, United States
- **Executive Consultant, Transportation Team** (Industry Experience) Jul 2018 – Nov 2019
Ernst & Young (EY) LLP, New Delhi, India
- **Visiting Undergraduate Research Student** May – Jul 2017
Purdue University, West Lafayette, United States
- **Summer Researcher — Purdue Undergraduate Research Experience (PURE)** May – Jul 2016
Lyles School of Civil Engineering, Purdue University, United States

RESEARCH INTERESTS

Transportation Engineering, Connected Vehicle Data Applications, Big Data Analytics, Connected and Autonomous Vehicles, Traffic Safety, Work Zone Safety, Applied Research, Traffic Shock Wave Theory, Scalable Performance Measures for Transportation Agencies.

PROFESSIONAL REGISTRATION

- **Professional Engineer (PE)**, Indiana License No. PE12600619
- Google Cloud Certified **Generative AI Leader**

SCIENTIFIC PUBLICATIONS

Google Scholar citations: **521** (Sept. 28, 2026); h-index 13 and i10-index 19

Theses

- **PhD Dissertation:** Integrating Connected Vehicle Data for Operational Decision Making (<https://doi.org/10.25394/PGS.22684822.v1>) Purdue University, April 2023.
Committee: Dr. Darcy Bullock, Dr. Samuel Labi, Dr. Konstantina Gkritza, Dr. James Krogmeier
- **Master's Thesis:** Reliable Corridor Level Travel Time Estimation using Probe Vehicle Data. Indian Institute of Technology (IIT) Madras, Chennai, May 2018.
Committee: Dr. V. Lelitha Devi, Dr. Gitakrishnan Ramadurai, Dr. Atul Narayan, Dr. A. Veeraragavan

Monograph & Patent

- **Sakhare, R. S.,** Desai, J., Mathew, J. K., McGregor, J., Kachler, M., & Bullock, D. M. (2024). Measuring and visualizing freeway traffic conditions: Using connected vehicle data (Joint Transportation Research Program No. TPF-5(514)). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317751> (528 downloads as of Sept. 28, 2026)
- (Patent) Devi, L. & **Sakhare, R. S.** (2024). System and Method for Estimating Reliable Corridor Level Travel Time Estimation Using Probe Vehicle Data (India Patent No. 529133). *The Patent Office, Government of India*.

Peer-Reviewed Journal Publications

- J32. **Sakhare, R.S.,** Desai, J., Gartner, C., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D.M. (2026). Deriving systemwide granular segment level traffic mobility performance metrics using connected vehicle data to identify improvement opportunities. *Transportation*. <https://doi.org/10.1007/s11116-026-10736-8>
- J31. **Sakhare, R.S.,** Desai, J., Overall, M., Mukai, J., Pava, J., McGregor, J., & Bullock, D.M. (2026). Work Zone Performance Measures Derived from Connected Vehicle Data for Safety and Mobility Assessment. *Future Transportation*, 6(1), 12. <https://doi.org/10.3390/futuretransp6010012>
- J30. Desai J.C., Gartner, C.M., **Sakhare, R.S.,** Cox E. & Bullock, D.M. (2026). Evaluating Safety Benefits of Ramp Metering by Leveraging Connected Vehicle Data. *IEEE Open Journal of Intelligent Transportation Systems*, 7, 190-199. <https://doi.org/10.1109/OJITS.2025.3650100>
- J29. Mukai, J., Desai, J., **Sakhare, R.S.,** Sturdevant, N. and Bullock, D. (2026) Quantifying Hard Breaking Events Occurring Prior to Secondary Crashes on Indiana Interstates. *Journal of Transportation Technologies*, 16, 150-165. <https://doi.org/10.4236/jtts.2026.161010>
- J28. Overall, M., Mukai, J., **Sakhare, R.S.,** Desai, J., Horton, D. and Bullock, D.M. (2026). Applications of Commercial Truck Dash Cameras for Work Zone Inspection and Monitoring. *Journal of Transportation Technologies*, 16, 95-114. <https://doi.org/10.4236/jtts.2026.161006>
- J27. Gartner, C.M., **Sakhare, R.S.,** Desai J.C., Sturdevant J. & Bullock, D.M. (2025). A Scalable Data Model for Signalized Intersection Performance Measures. *IEEE Access*, 13, 197851-197863. <https://doi.org/10.1109/ACCESS.2025.3635007>
- J26. Desai, J., **Sakhare, R. S.,** McGregor, J., Snyder, B., & Bullock, D. M. (2025). Evaluating Impact of Automated Work Site Speed Enforcement Using Connected Vehicle Data. *Applied Sciences*, 15(22), 12140. <https://doi.org/10.3390/app152212140>
- J25. Mathew, J., Desai, J., **Sakhare, R. S.,** Hunter, J. and Bullock, D. (2025) Spatiotemporal Analysis of Pavement Roughness Using Connected Vehicle Data for Asset Management. *Journal of Transportation Technologies*, 15, 1-16. <https://doi.org/10.4236/jtts.2025.151001>
- J24. Overall, M. W., Mukai, J., **Sakhare, R. S.,** Desai, J., Lowther, H., & Bullock, D. M. (2025). Measuring the Impact of Recovery Resource Delay on Traffic Incident Management Clearance Times. *Future Transportation*, 5(4), 171. <https://doi.org/10.3390/futuretransp5040171>
- J23. Driscoll, T., **Sakhare, R.S.,** Desai, J. and Bullock, D.M. (2025) Utilizing Connected Vehicle Data to Identify Impacts of Congestion on Adjacent Agencies. *Journal of Transportation Technologies*, 15, 522-541. <https://doi.org/10.4236/jtts.2025.154024>
- J22. **Sakhare, R.S.,** Desai, J., Mathew, J.K. and Bullock, D.M. (2024). Assessing the Interstate Coverage of Commercial Trucks Capable of Providing Roadway Imagery via On-Vehicle Dash Camera in the United States. *IEEE Access*, vol. 12, pp. 173517-173529. <https://doi.org/10.1109/ACCESS.2024.3503368>
- J21. **Sakhare, R.S.,** Desai, J., Mathew, J. and Bullock, D.M. (2024). Impact of 2024 Solar Eclipse on National Traffic Mobility using Connected Vehicle Data and Images. *Transportation Research Interdisciplinary Perspectives*, 27. <https://doi.org/10.1016/j.trip.2024.101225>

- J20. Brinster, G. L., Desai, J., **Sakhare, R. S.**, Mathew, J. K., & Bullock, D. M. (2024). Expanding on methodologies for analyzing truck speeds in states with differential speed limits. *Safety*, 10(4), 99. <https://doi.org/10.3390/safety10040099>
- J19. Vajpayee, V., Saldivar-Carranza, E. D., **Sakhare, R. S.**, & Bullock, D. M. (2024). Large Scale Evaluation of Normalized Hard-Braking Events Derived from Connected Vehicle Trajectory Data at Signalized Intersections, Roundabouts, and All-Way Stops. *Future Transportation*, 4(3), 968-984. <https://doi.org/10.3390/futuretransp4030046>
- J18. Brinster, G.L., Desai, J., Overall, M.W., Gartner, C., **Sakhare, R.S.**, Mathew, J.K., Evans, N. and Bullock, D. (2024) Evaluating the Robustness of MDSS Maintenance Forecasts Using Connected Vehicle Data. *Journal of Transportation Technologies*, 14, 549-569. <https://doi.org/10.4236/jtts.2024.144030>
- J17. Saldivar-Carranza, E. D., **Sakhare, R. S.**, Desai, J., Mathew, J. K., Sivakumar, A. J., Mukai, J., & Bullock, D. M. (2024). Impact of Privacy Filters and Fleet Changes on Connected Vehicle Trajectory Datasets for Intersection and Freeway Use Cases. *Smart Cities*, 7(5), 2366-2391. <https://doi.org/10.3390/smartcities7050093>
- J16. Desai, J., Saldivar-Carranza, E. D., **Sakhare, R. S.**, Mathew, J. K., & Bullock, D. M. (2024). Impact of In-Cab Alerts on Connected Truck Speed Reductions in Indiana. *Vehicles*, 6(4), 1857-1871. <https://doi.org/10.3390/vehicles6040090>
- J15. **Sakhare, R. S.**, Desai, J., Saldivar, E. D., & Bullock, D. M. (2023). Methodology for Monitoring Border Crossing Delays with Connected Vehicle Data: United States and Mexico Land Crossings Case Study. *Future Transportation*, 4(1), 107-129. <https://doi.org/10.3390/futuretransp4010007>
- J14. **Sakhare, R. S.**, Li, H., & Bullock, D. M. (2023). Methodology for the Identification of Shock Wave Type and Speed in a Traffic Stream Using Connected Vehicle Data. *Future Transportation*, 3(4), 1147–1174. <https://doi.org/10.3390/futuretransp3040063>
- J13. **Sakhare, R. S.**, Zhang, Y., Li, H., & Bullock, D. M. (2023). Impact of Rain Intensity on Interstate Traffic Speeds Using Connected Vehicle Data. *Vehicles*, 5(1), 133–155. <https://doi.org/10.3390/vehicles5010009>
- J12. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2023). Analysis of Connected Vehicle Data to Quantify National Mobility Impacts of Winter Storms for Decision Makers and Media Reports. *Future Transportation*, 3(4), 1292-1309. <https://doi.org/10.3390/futuretransp3040071>
- J11. Desai, J., Mathew, J., Li, H., Mukai, J., **Sakhare, R. S.**, & Bullock, D. M. (2023). Correlating Connected Vehicle Estimated Deceleration Events with Crash Incidents near Interstate Interchanges. *Journal of Transportation Technologies*, 13, 674-688. <https://doi.org/10.4236/jtts.2023.134031>
- J10. **Sakhare, R. S.**, Hunter, M., Mukai, J., Li, H., & Bullock, D. M. (2022). Truck and Passenger Car Connected Vehicle Penetration on Indiana Roadways. *Journal of Transportation Technologies*, 12, 578-599. <https://doi.org/10.4236/jtts.2022.124034>
- J9. **Sakhare, R. S.**, Desai, J., Li, H., Kachler, M. A., Bullock, D. M. (2022). Methodology for Monitoring Work Zones Traffic Operations Using Connected Vehicle Data. *Safety*, 8(2), 41. **(Editor's Choice Award)** <https://doi.org/10.3390/safety8020041>
- J8. Mahlberg, J. A., Desai, J., Li, H., **Sakhare, R. S.**, Wells, T., Bullock, D. M. (2022). Calibration and Confidence in Snowplow Fleet Operations and Fleet Telematics. *Journal of Transportation Technologies*, 12(4), 696-710. <https://doi.org/10.4236/jtts.2022.124040>
- J7. **Sakhare, R. S.**, Desai, J. C., Mahlberg, J., Mathew, J. K., Kim, W., Li, H., McGregor, D., & Bullock, D. M. (2021). Evaluation of the Impact of Queue Trucks with Navigation Alerts Using Connected Vehicle Data. *Journal of Transportation Technologies*, 11(4), 561-576. <https://doi.org/10.4236/jtts.2021.114035> **(Recognized by ITS Midwest as the 2021 Project of the Year)**
- J6. **Sakhare, R. S.**, Desai, J. C., Mathew, J. K., McGregor, J. D., & Bullock, D. M. (2021). Evaluation of the impact of presence lighting and digital speed limit trailers on interstate speeds in Indiana work zones. *Journal of transportation technologies*, 11(2), 157-167. <https://doi.org/10.4236/jtts.2021.112010>
- J5. Mathew, J. K., Desai, J. C., **Sakhare, R. S.**, Kim, W., Li, H., & Bullock, D. M. (2021). Big data applications for managing roadways. *Institute of Transportation Engineers. ITE Journal*, 91(2), 28-35.
- J4. Mahlberg, J. A., **Sakhare, R. S.**, Li, H., Mathew, J. K., Bullock, D. M., & Surnilla, G. C. (2021). Prioritizing Roadway Pavement Marking Maintenance Using Lane Keep Assist Sensor Data. *Sensors*, 21(18), 6014. <https://doi.org/10.3390/s21186014>
- J3. Desai, J., Mahlberg, J., Kim, W., **Sakhare, R. S.**, Li, H., McGuffey, J., & Bullock, D. M. (2021). Leveraging telematics for winter operations performance measures and tactical adjustment. *Journal of Transportation Technologies*, 11(4), 611-627. <https://doi.org/10.4236/jtts.2021.114038>

- J2. **Sakhare, R. S.** & Vanajakshi, L. (2020). Reliable Corridor Level Travel Time Estimation Using Probe Vehicle Data. *Transportation Letters*, 12:8, 570-579. <https://doi.org/10.1080/19427867.2019.1671041>
- J1. **Sakhare, R. S.**, Dev, L. K., Prasad, K. E., & Robinson, R. G. (2016). A Double Acting Piston Based Automatic Volume Change Apparatus. *Geotechnical Testing Journal*, 39(6), 899-905. <http://doi.org/10.1520/GTJ20150296> (IGS Best Paper Award)

Key Project / Technical Reports

- T22. Thompson, A., Desai, J. C., **Sakhare, R. S.**, & Bullock, D. M. (2026). Research support to INDOT on systemwide asset condition assessment using connected vehicle data (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2026/04). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318612>
- T21. **Sakhare, R. S.**, Desai, J., Gartner, C., Mukai, J., Sturdevant, N., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D. M. (2025). Connected vehicle trajectory data to screen network for hard-braking and hard-acceleration events (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/30). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318579>
- T20. Desai, J., **Sakhare, R. S.**, McGregor, J., Sturdevant, J., & Bullock, D. M. (2025). Worksite speed control system (WCS) (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/33). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318595>
- T19. Overall, M. W., Mukai, J., Desai, J., **Sakhare, R. S.**, Kachler, M., McGregor, J., & Bullock, D. M. (2025). Continued deployment of Indiana work zone analytics (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/31). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318581>
- T18. Desai, J. C., Gartner, C., **Sakhare, R. S.**, Cox, E. D., Sturdevant, J., & Bullock, D. M. (2025). Research support to INDOT on I-465 Southeast variable speed limit and ramp meter project (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/32). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318592>
- T17. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). Case studies documenting lessons learned and identifying opportunities for maintenance of traffic (MOT) improvements (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/29). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318578>
- T16. Desai, J., **Sakhare, R. S.**, Mathew, J. K., Sturdevant, N., Cox, E., & Bullock, D. M. (2025). Communication of fixed and mobile warnings to commercial trucks using in-cab notification (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/24). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318232>
- T15. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Lowther, H., Cox, E., Sturdevant, J., & Bullock, D. M. (2025). Development of incident management performance measures database and supporting training material (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/21). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317907>
- T14. **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). Crawfordsville District interstate speed profiles: April 2020–March 2025. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317852> (1,888 downloads as of 2/26/2026)
- T13. **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). *Indiana interstate speed profiles 2020–2024*. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317843>
- T12. Gayen, S., Saldivar-Carranza, E. D., **Sakhare, R. S.**, Desai, J., Mathew, J. K., Shellhamer, N., Ruble, T., McCoy, D., Gallagher, J., & Bullock, D. M. (2024). Statewide screening of signalized intersections for capacity improvements (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2024/20). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317755>
- T11. Brinster, G. L., Desai, J., Overall, M. W., Gartner, C., **Sakhare, R. S.**, Mathew, J. K., Evans, N., & Bullock, D. M. (2024). Evaluating the robustness of MDSS forecast and compliance with recommendations (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2024/31). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317805>
- T10. **Sakhare, R. S.**, Desai, J., Woker, W., Li, H., Mathew, J., Mahlberg, J., Saldivar-Carranza, E. D., Horton, D., & Bullock, D. M. (2023). Connected vehicle centric dashboards for TMC of the future (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/17). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317642>

- T9. Desai, J., **Sakhare, R. S.**, Mahlberg, J., Mathew, J. K., Li, H., & Bullock, D. M. (2023). Implementation of enhanced probe data (CANBUS) for tactical workzone and winter operations management (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/18). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317643>
- T8. Mahlberg, J. A., Desai, J., Malackowski, H., Li, H., Mathew, J. K., **Sakhare, R. S.**, Wells, T., & Bullock, D. M. (2023). Crowdsourcing/winter operations dashboard upgrade (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/21). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317652>
- T7. **Sakhare, R. S.**, Mahlberg, J., Mathew, J. K., Desai, J., Li, H., Bullock, D. M. (2022). Impacts to traffic behavior from queue warning truck: Current pilot project (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2022/26). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317448>
- T6. Mathew, J. K., Li, H., Desai, J., **Sakhare, R. S.**, Saldivar-Carranza, E., Hunter, M., Scholer, B., & Bullock, D. M. (2022). Integration of probe data tools into TMC operations (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2022/01). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317363>
- T5. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2022). National mobility analysis for all interstate routes in the United States: December 2022. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317591>
- T4. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2022). National mobility analysis for all interstate routes in the United States: August 2022. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317585>
- T3. **Sakhare, R. S.**, Li, H., Mathew, J. K., Desai, J., Horton, D., & Bullock, D. M. (2022). Indiana interstate speed profiles 2018–2022. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317589>
- T2. **Sakhare, R. S.**, Desai, J., Mathew, J. K., Kim, W., Mahlberg, J., Li, H., & Bullock, D. M. (2021). Evaluating the impact of vehicle digital communication alerts on vehicles (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2021/19). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317324>
- T1. Day, C. M., McNamara, M. L., Li, H., **Sakhare, R. S.**, Desai, J., Cox, E. D., Horton, D. K., & Bullock, D. M. (2016). 2015 Indiana mobility report and performance measure dashboards. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284316352>

Peer-Reviewed Conference & Workshop Presentations

- C26. **Sakhare, R.S.**, Desai, J., Overall, M., Mukai, J., Pava, J., McGregor, J., & Bullock, D. M. (2026, January). Improving Vehicle and Worker Safety in Dynamic Work Zones – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02289). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24853>
- C25. Desai, J., Gartner, C., **Sakhare, R.S.**, Cox, E., & Bullock, D. M. (2026, January). Improving Vehicle and Worker Safety in Dynamic Work Zones – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02345). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24853>
- C24. **Sakhare, R.S.**, Desai, J., Gartner, C., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D. M. (2026, January). Travel Data and Networks: Collection, Modeling, and Mobility Insights – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-01332). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24601>
- C23. Gartner, C., **Sakhare, R.S.**, Desai, J., Sturdevant, J., & Bullock, D. M. (2026, January). Advanced Modeling, Optimization, and Safety for Signalized Intersections – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02156). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24854>
- C22. Desai, J., **Sakhare, R.S.**, McGregor, J., Snyder, B., & Bullock, D. M. (2026, January). Data-Driven Approaches for Perception, Prediction, and Decision Making in Autonomous Vehicles – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02221). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24522>
- C21. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Horton, D., & Bullock, D. M. (2026, January). Modeling Traffic Speed and Network Performance Under Disruptive Events – Poster Session. In Transportation Research

- Board 105th Annual Meeting (No. TRBAM-26-03560). TRB.
<https://annualmeeting.mytrb.org/OnlineProgram/Details/24845>
- C20. Driscoll, T., **Sakhare, R.S.**, Desai, J., & Bullock, D. M. (2026, January). Emerging Research Topic in Strategic Management and Organizational Excellence – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03893). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24765>
- C19. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Hillary, L., & Bullock, D. M. (2026, January). Frameworks for Connected Traffic and Emergency Operations Management – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03559). TRB.
<https://annualmeeting.mytrb.org/OnlineProgram/Details/24533>
- C18. Mathew, J., Desai, J., **Sakhare, R.S.**, Hunter, J., & Bullock, D. M. (2025, January). Innovative Use of Emerging Data and Criteria for Maintenance Management – Lectern Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-01326). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22915>
- C17. Brinster, G., Desai, J., Overall, M., Gartner, C., **Sakhare, R.S.**, Mathew, J., Evans, N., & Bullock, D. M. (2025, January). Innovative Use of Emerging Data and Criteria for Maintenance Management – Lectern Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-02443). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22915>
- C16. **Sakhare, R.S.**, Desai, J., Mathew, J., & Bullock, D. M. (2025, January). Statewide and National Transportation Data Research – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-03615). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22967>
- C15. Bullock, D. M., & **Sakhare, R.S.** (2025, January). State Department of Transportation Research: High-Value Project Winners – Poster Session. In Transportation Research Board 104th Annual Meeting (No. P25-20182). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22727>
- C14. Saldivar-Carranza, E., **Sakhare, R.S.**, Desai, J., Mathew, J., Sivakumar, A. J., Mukai, J. & Bullock, D. M. (2025, January). Intelligent Transportation Systems in the Era of Connected vehicles and Vehicle to Everything Communication – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-03624). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22967>
- C13. Desai, J., Saldivar-Carranza, E., **Sakhare, R.S.**, Mathew, J., & Bullock, D. M. (2025, January). Toward Better Driving: Innovations for Improved Driving Performance and Enhanced Safety – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-02784). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22931>
- C12. Desai, J., Mathew, J., Li, H., **Sakhare, R.S.**, Horton, D., & Bullock, D. M. (2024, January). Statewide and National Data and Information Systems – Poster Session. In Transportation Research Board 103rd Annual Meeting (No. TRBAM-24-04421). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/21103>
- C11. **Sakhare, R.S.**, Zhang, Y., Li, H., & Bullock, D. M. (2023, January). Road Weather Management Solutions for Improved Safety and Operations – Lectern Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-02713). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19287>
- C10. Mahlberg, J., Desai, J., Li, H., **Sakhare, R.S.**, Wells, T., & Bullock, D. M. (2023, January). Advancements Available to Maintenance Equipment Fleets Within Departments of Transportation and Other Fleet Operations – Lectern Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-01921). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19002>
- C9. **Sakhare, R.S.**, Hunter, M., Mukai, J., Li, H., & Bullock, D. M. (2023, January). Road Scholars: New Research in Travel Time, Speed, and Reliability Data – Poster Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-01814). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19057>
- C8. Desai, J., Mahlberg, J., Woosung, K., **Sakhare, R.S.**, Li, H., McGuffey, J., & Bullock, D. M. (2022, January). Advancements in Winter Maintenance – Lectern Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-02228). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17477>
- C7. **Sakhare, R.S.**, Desai, J., Mahlberg, J., Mathew, J., Woosung, K., Li, H., McGregor, J., & Bullock, D. M. (2022, January). Freeway and Regional Transportation Systems and Operations – Poster Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-02130). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17526>

- C6. Mahlberg, J., **Sakhare, R.S.**, Li, H., Mathew, J., Bullock, D. M. & Surnilla, G. (2022, January). Traffic Control Devices – Poster Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-01292). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17477>
- C5. Desai, J., **Sakhare, R. S.**, Rogers, S., Mathew, J. K., Habib, A., & Bullock, D. (2021, September). Using Connected Vehicle Data to Evaluate Impact of Secondary Crashes on Indiana Interstates. In 2021 IEEE International Intelligent Transportation Systems Conference (ITSC) (pp. 4057-4063). IEEE. <https://doi.org/10.1109/ITSC48978.2021.9564653>
- C4. **Sakhare, R. S.**, Desai, J. C., Li, H., Mathew, J. K., & Bullock, D. M. (2021, January). Predicting Weather Impact on Traffic Speeds Using the Multi-Radar Multi-Sensor System. In 101st American Meteorological Society Annual Meeting. AMS. <https://ams.confex.com/ams/101ANNUAL/meetingapp.cgi/Paper/382590>
- C3. **Sakhare, R. S.**, Mathew, J. K., Avr, A., Hubbard, S. M., Devi, L., & Bullock, D. M. (2018, January). Comparison of Bluetooth and Bus GPS Data for Estimating Arterial Travel Time and Trip Chaining. In Transportation Research Board 97th Annual Meeting (No. TRBAM-18-01695). TRB. <https://trid.trb.org/View/1495014>
- C2. Li, H., **Sakhare, R. S.**, Mathew, J. K., Mackey, J., & Bullock, D. M. (2018, January). Estimating Intersection Control Delay Using High Fidelity Commercial Probe Vehicle Trajectory Data. In Transportation Research Board 97th Annual Meeting (No. TRBAM-18-01695). TRB. <https://trid.trb.org/View/1494388>
- C1. McNamara, M., **Sakhare, R. S.**, Li, H., Baldwin, M., & Bullock, D. M. (2017, January). Integrating Crowdsourced Probe Vehicle Traffic Speeds into Winter Operations Performance Measures. In Transportation Research Board 96th Annual Meeting (No. TRBAM-17-00161). TRB. <https://trid.trb.org/View/1437147>

Key Technical Presentations

- P9. **Sakhare, R.S.**, & Taylor, R. (2026, March). Segment Mobility and Signal Performance: Leveraging CV Metrics – Lectern Session. In 112th Purdue Road School (No. 131).
- P8. **Sakhare, R.S.**, Thompson, A., & Sturdevant, N. (2026, March). Evaluation of Asset Identification Technology Utilizing Truck Images – Lectern Session. In 112th Purdue Road School (No. 159).
- P7. **Sakhare, R.S.**, & McGregor, J. (2026, March). Utilizing CV Data and Truck Imagery for Roadway Monitoring – Lectern Session. In 112th Purdue Road School (No. 159).
- P6. Gartner, C., & **Sakhare, R.S.** (2026, February). Connected Vehicle Data Use Cases for Practitioners – Lectern Session. In 2026 ITE Indiana Section February Tech Lunch. <https://indianaite.starchapter.com/meetinginfo.php?id=202&ts=1768590344>
- P5. **Sakhare, R.S.** (2026, February). Work Zone Performance Measures Derived from Connected Vehicle Data that Map into the FHWA Rule on Work Zone Safety and Mobility – Seminar. In Purdue ITE Weekly Seminar Series.
- P4. **Sakhare, R.S.** (2023, April). Impact of Rain Intensity on Interstate Traffic Speeds Using Connected Vehicle Data – Seminar. In Purdue ITE Weekly Seminar Series.
- P3. **Sakhare, R.S.**, & Hillary, L. (2023, March). Identifying Traffic Incident Management Event Sequence using CV Data, Crash Reports, and ITS Images – Lectern Session. In 109th Purdue Road School (No. 61).
- P2. **Sakhare, R.S.** (2023, March). Automation System Performance Issues/Truck and Car CV Penetration – Lectern Session. In 109th Purdue Road School (No. 117).
- P1. **Sakhare, R.S.**, & Desai, J. (2023, March). Methodology for Monitoring Work Zones Traffic Operations using CV Data: Implementation on Indiana Interstate with UAS Imagery – Lectern Session. In 109th Purdue Road School (No. 182).

FUNDED RESEARCH PROJECTS

Roles - PI: Principal Investigator; RE: Research Engineer; GRA: Graduate Research Assistant

Funding Organizations - FHWA: Federal Highway Administration; INDOT: Indiana Department of Transportation; CCAT: Center for Connected and Automated Transportation.

- **FHWA** — TPF-5(514): Work Zone Analytics (RE, 2023-2026, \$0.89M)
- **FHWA** — TPF-5(519): Expansion: Enhanced Traffic Signal Performance Measures (RE, 2023-2026, \$1.3M)
- **INDOT** — SPR-5005: Evaluation of Asset Identification Technology (**Co-PI, 2025-2027**, \$301,966)
- **INDOT** — SPR-5021: Traffic Signal Freight Prioritization via Vehicle to Infrastructure (V2I) Communications (**Co-PI, 2026-2027**, \$383,966)

- **INDOT** — SPR-5020: Identifying Locations with Abnormally High Wrong Way Driving or Interstate U-Turns (Co-PI, 2025-2027, \$224,109)
- **INDOT** — SPR-4928: Connected Vehicle Trajectory Data to Screen Network for Hard Braking and Hard Acceleration Events (Co-PI, 2024-2025, \$218,390)
- **INDOT** — SPR-4933: Worksite Speed Control System (WSCS) (RE, 2024-2025)
- **INDOT** — SPR-4907: Systemwide Asset Condition Assessment using Connected Vehicle Data (RE, 2024-2025)
- **INDOT** — SPR-4857: Statewide Screening of Signalized Intersections for Capacity Improvements (RE, 2023-2024)
- **INDOT** — SPR-4854: MOT for MOT (RE, 2023-2025)
- **INDOT** — SPR-4851: Development of Incident Management Performance Measure Database and IN Time Training Material (RE, 2023-2025)
- **INDOT** — SPR-4850: Continued Deployment of Indiana Work Zone Analytics (RE, 2023-2025)
- **INDOT** — SPR-4803: Communication of Fixed and Mobile Warning to Commercial Trucks Using in Cab Notification (RE, 2023-2025)
- **INDOT** — SPR-4704: Evaluating the Robustness of MDSS Forecast and Compliance with Recommendations (RE, 2022-2024)
- **INDOT** — SPR-4639: Connected Vehicle Centric Dashboards for TMC of the Future (GRA, 2021-2023)
- **INDOT** — SPR-4636: Research Support to INDOT on I-465 Southeast Variable Speed Limit and Ramp Meter Project (RE, 2021-2025)
- **INDOT** — SPR-4603: Crowdsourcing/Winter Operations Dashboard Upgrade (GRA, 2021-2023)
- **INDOT** — SPR-4536: Implementation of Enhanced Probe Data (CANBUS) for Tactical Workzone and Winter Operations Management (GRA, 2020-2023)
- **CCAT** — Methodologies for Monitoring Work Zone Performance Measures across State Border Highways using Connected Vehicle Data (RE, 2023-2025)
- **INDOT** — SPR-4600: Impacts to Traffic Behavior from Queue Warning Truck – Current Pilot Project (GRA, 2021-2022)
- **INDOT** — SPR-4524: Evaluation of the Impact of On Vehicle Digital Communication Alerts to Improve INDOT and Motorist Safety (GRA, 2020-2022)
- **INDOT** — SPR-4451: Integration of Probe Data Tools into TMC Operations (GRA, 2020-2022)

TEACHING & MENTORING EXPERIENCE

- **Guest Lecture** for CE36100 Transportation Engineering, Purdue University on topic of *Connected Vehicle Data and its Applications* (April 29, 2026)
- **Invited Speaker** at the Transportation division, Department of Civil Engineering, IIT Madras on *Methodology for the Identification of Shock Wave Type and Speed in a Traffic Stream Using Connected Vehicle Data*; Interaction with undergraduate and graduate student in Civil Engineering. (August 21, 2024)
- **Instructor** for Mobility Course and Workshop SPARC (Scheme for Promotion of Academic and Research Collaboration, Government of Indian Initiative) - Developed course material and taught 3 session modules (2021)
- **Teaching Assistant** to Dr. Atul Narayan for CE2080 Civil Engineering Surveying course Department of Civil Engineering, IIT Madras (2017-2018)
- **Courses Prepared to Teach:** Transportation Engineering; Big Data Analytics in Transportation; Traffic Operations

Graduate Student Research Co-Advising/Mentorship

- **Myles Overall, Ph.D. in Civil Engineering** — Incident Management Performance Measures (SPR-4851)
- **Chris Gartner, M.S. in Civil Engineering** — Statewide performance measures (SPR-4857)
- **Andrew Thompson, M.S. in Electrical and Computer Engineering** — Road Asset Identification (SPR-4907)
- **Justin Mukai, M.S. in Civil Engineering** — Secondary crashes at the back of queue (J29)
- **Thomas Driscall, M.S. in Mechanical Engineering** — Cross-border queueing (J23)

- **Neelesh Gopalkrishnan, M.S. in Computer Science** — CV Penetration dashboard
- **Ahsmitha Sivakumar, B.Tech. in Civil Engineering** — Toll Plaza performance measures
- **Anirud Nandakumar, B.Tech. in Civil Engineering** — Normalized Hard braking evaluations

HONORS, AWARDS & ACHIEVEMENTS

- **2024** — Selected for the Google Cloud Research Innovator as part of the [fourth cohort](#)
- **2023** — Edward J. Cox Memorial Scholarship, Indiana ITE Section
- **2023** — Best Student Speaker Award, Purdue ITE
- **2021** — ITS Midwest Project of the Year Award Winner
- **2021 & 2022** — Elevator Pitch Event Winner, Purdue ITE
- **2021** — Student Team Design Competition Winner, Great Lakes District ITE
- **2020** — International Collegiate Traffic Bowl Championship Winner, ITE International
- **2020** — Best Poster Presentation Sigma Xi Scientific Society, Purdue University
- **2020-2023** — Christopher B. & Susan S. Burke Graduate Program Research Assistantship, Purdue University
- **2018** — Exemplary and All-round Best Performance in Dual Degree Program, Civil Engineering, Indian Institute of Technology (IIT) Madras, India
- **2018** — Winner at CEA Technical Events (Case Study, Prabandha, Bon Auto and Aquanomics), IIT Madras, Chennai, India
- **2017** — Invited Visiting Undergraduate Student at Purdue University
- **2016** — Gold Medal in Civil Engineering, National Design and Research Forum, Institute of Engineers, India
- **2016** — Mr. H. C. Verma Diamond Jubilee Awards, Indian Geotechnical Society (IGS), India
- **2016** — Purdue Undergraduate Research Experience (PURE) Program Scholar; Among the top 3 Civil Engineering students chosen across India
- **2016** — Represented India (Team of four) at Engineering Case Competition in Thailand organized by the Faculty of Chulalongkorn University
- **2015** — Winner of Smart City Challenge Vellore-Engineers for Life Program
- **2015** — Successfully Completed the Engineers for Life Program, AidVision Initiative
- **2014** — Village's Choice Award, Faculty of Engineering, Chulalongkorn University, Thailand
- **2014** — Represented India at the Engineering Case Competition (ECC) in Thailand
- **2013** — Ranked top 0.2 percentile (2,445th) of 1.45 million students at IIT Joint Entrance Exam; considered amongst world's toughest examinations

ACADEMIC SERVICE

- **2025 Aug., - Current** — Transportation Research Board (TRB) **Committee Member** ACF13, Standing Committee on Limited Access Roadway Operations
- **2024, 2025 & 2026** — **Planning Committee Member** Purdue Road School; 112 year old annual conference
- **2023, 2024 & 2025** — **Session Moderator** Safety track (2023) Project Management/Planning track (2024) Project Management/Planning track (2025) at Purdue Road School
- **2022** — Institute of Transportation Engineers (ITE) **Executive Board Member – Purdue Chapter**; Received *ITE Student Chapter Award* from ITE International; Received *Outstanding Student Chapter from ITE Great Lakes*
- **2024** — Judge for Spring Undergraduate Research Conference
- Active Reviewer for following Journals:
 - Transportation Research Part C: Emerging Technologies
 - Journal of Transportation Engineering, Part A: Systems, ASCE
 - Transportation Letters: the International Journal of Transportation Research
 - Transportation Research Board
 - Transportation Research Records

- Transportation Research Interdisciplinary Perspectives
- Journal of the Institution of Engineers (India): Series A
- ITS World Congress
- Transportation in Developing Economies
- International Journal of Digital Earth

COMMUNITY LEADERSHIP & EXTRACURRICULARS

- **2022** — National Collegiate Cricket Association's (NCCA) Midwest Zone Championship Winner (Captain of the Purdue Cricket team)
- **2022** — Midwest Cricket Tournament Runner-up (more than 60 participating teams)
- **2022 & 2023** — Purdue Indoor Cricket League Winner (Captain of the team in 2023)
- **2018** — Winner in CEA Cricket Tennis Ball Tournament (runner up in 2017)
- **2017** — Fourth in Men's Cricket, IITM Institute Gymkhana Intramural Competitions
- **2016** — Head Member at International & Alumni Relations Student Council, IIT Madras
- **2016** — Volunteer for Annual Terry Fox Run, The Terry Fox Foundation, Chennai
- **2015** — Head Member at Civil Engineering Association (CEA), IIT Madras (2015)
- **2015** — Student Mentor, MITR Office of the Dean Students, IIT Madras
- **2015** — Ambience & Design Super-Coordinator at Shaastra, Technical Event, IIT Madras
- **2014** — Events & Projects Coordinator at Sustainability Network, IIT Madras
- **2010** — Competed at State Level Archery Competitions, Maharashtra, India

SELECT MEDIA COVERAGE

- **2024, Dec. 24** — The New York Times, How Your Car Might Be Making Roads Safer. <https://www.nytimes.com/2024/12/20/technology/connected-cars-roads-data.html>
- **2023, Apr. 04** — The Times of India, Explained: How connected vehicle data can help track and predict traffic congestion during storm. <https://timesofindia.indiatimes.com/videos/international/explained-how-connected-vehicle-data-can-help-track-and-predict-traffic-congestion-during-storms/videoshow/99244161.cms>
- **2023, Mar. 22** — Purdue News, Connected vehicles the latest tool to give engineers real-time insight into highway traffic congestion issues. <https://www.purdue.edu/newsroom/2023/Q1/connected-vehicles-the-latest-tool-to-give-engineers-real-time-insight-into-highway-traffic-congestion-issues>
- **2021, Apr. 07** — Google Cloud Blog, Purdue University traffic research program cuts data analysis and batching from hours to minutes with BigQuery. <https://cloud.google.com/customers/purdueitrp>
- **2021, Apr. 05** — Medium, Putting data in the driver's seat. <https://medium.com/purdue-engineering/putting-data-in-the-drivers-seat-e06122696fc8>
- **2021, Sept. 17** — HAAS Alert *Industry News*, Digital Alerting Helps Reduce Hard Braking by 80% in Purdue University Study. <https://www.haasalert.com/report/purdue-study>

SCHOLARLY PROFILES

- **Google Scholar:** <https://scholar.google.com/citations?user=4crwCDoAAAAJ&hl=en>
- **Research Gate:** <https://www.researchgate.net/profile/Rahul-Suryakant-Sakhare>
- **ORCID:** <https://orcid.org/0000-0001-7843-5707>
- **LinkedIn:** <https://www.linkedin.com/in/rahulsakhare/>